Laboratory Protocol: Agarose Gel Electrophoresis of DNA

BIO 210 — Cell Biology Laboratory · Lab 4

Purpose

Agarose gel electrophoresis separates DNA fragments by size. An electric field drives negatively charged DNA through a porous agarose matrix toward the positive electrode; smaller fragments move faster and travel farther than larger ones. By comparing your sample bands to a DNA ladder of known sizes, you can estimate the length of unknown fragments and confirm whether a procedure — such as a restriction digest or a PCR — worked as expected.

In this lab you will pour an agarose gel, load DNA samples mixed with loading dye, run the gel, and photograph the results under a blue-light or UV transilluminator. You will then interpret the band pattern.

Learning Objectives

By the end of this lab you should be able to:

- Explain why DNA migrates toward the positive electrode and why smaller fragments move faster.
- Prepare an agarose gel of an appropriate percentage for a target fragment size.
- Load samples correctly and run a gel at a safe, effective voltage.
- Use a DNA ladder to estimate fragment sizes.
- Identify the most common sources of error and how to avoid them.

Background

DNA is uniformly negatively charged because of its phosphate backbone, so in an electric field every fragment moves toward the anode (the positive electrode). The agarose matrix acts as a molecular sieve: long fragments become entangled and move slowly, while short fragments thread through the pores quickly. The relationship between fragment size and migration distance is roughly logarithmic over the useful range of a gel.

The **percentage of agarose** sets the pore size and therefore the range of fragment sizes the gel resolves well:

Agarose concentration	Optimal resolution range
0.7%	1,000 to 12,000 base pairs (large fragments)
1.0%	500 to 10,000 base pairs (general purpose)
2.0%	100 to 2,000 base pairs (small fragments)

For this lab we use a **1.0% gel**, which resolves the fragment sizes in our samples well.

Materials and Reagents

Equipment

- Horizontal gel electrophoresis tank with casting tray and well comb
- Power supply
- Microwave or hot plate for melting agarose
- Micropipettes (2–20 µL range) and tips
- Blue-light or UV transilluminator with camera or gel-documentation system
- Microcentrifuge tubes and a tube rack

Reagents

- Agarose powder, molecular biology grade
- 1X TAE buffer (Tris-acetate-EDTA) — used both to make the gel and to fill the tank
- DNA stain — **SYBR Safe** or an equivalent low-hazard stain (preferred over ethidium bromide for teaching labs)
- 6X DNA loading dye
- DNA ladder (molecular-weight marker) of known fragment sizes
- DNA samples to be analyzed

Safety

Read this section before beginning. The instructor will confirm you understand it before issuing reagents.

- **Molten agarose is hot enough to cause burns.** Wear heat-resistant gloves when handling the flask after microwaving, swirl gently away from your face, and let the solution cool to about 55 °C (comfortable to hold against the wrist) before pouring. Superheated agarose can boil over suddenly when swirled.

- **DNA stain.** Even low-hazard stains such as SYBR Safe are handled with gloves and disposed of as directed. **Ethidium bromide, if used, is a mutagen** and requires nitrile gloves, dedicated waste containers, and instructor supervision — this lab uses SYBR Safe specifically to reduce that hazard.
- **The transilluminator.** A UV transilluminator emits ultraviolet light that damages eyes and skin. **Never look directly at an operating UV source and never expose bare skin to it** — use the safety shield, a face shield, and gloves. Blue-light transilluminators are safer but still warrant eye protection.
- **The power supply produces voltages capable of delivering a dangerous shock.** Connect leads before powering on, never open the tank lid while the power is on, and turn the supply off before removing the gel.
- Wear gloves, eye protection, and a lab coat throughout. Wash hands when finished.

Procedure

Part 1 — Cast the gel

1. Measure 0.5 g of agarose into a flask and add 50 mL of 1X TAE buffer to make a 1.0% gel.

2. Microwave in short bursts, swirling between bursts, until the agarose is fully dissolved and the solution is clear with no floating particles. Use heat-resistant gloves.

3. Allow the solution to cool to approximately 55 °C.

4. Add DNA stain (SYBR Safe) to the cooled agarose at the concentration specified on the reagent label, and swirl gently to mix without creating bubbles.

5. Seal the casting tray, insert the well comb, and pour the molten agarose into the tray. Pop any bubbles with a pipette tip.

6. Let the gel solidify for 20 to 30 minutes until it is opaque and firm.

Part 2 — Prepare and load samples

7. Place the set gel (still in its tray) into the electrophoresis tank and add 1X TAE buffer until it just covers the gel surface. **Use the same 1X TAE buffer you used to make the gel** — mismatched buffers distort migration.

8. Gently remove the comb to reveal the wells.

9. For each sample, combine 5 µL of DNA with 1 µL of 6X loading dye in a microcentrifuge tube and mix. The loading dye adds density so the sample sinks into the well and provides tracking dyes that show how far the run has progressed.

10. Load the DNA ladder into the first well.

11. Load each prepared sample into a separate well, recording which sample is in which lane. Pipette slowly so the dense sample settles into the well rather than drifting out.

Part 3 — Run the gel

12. Place the lid on the tank, matching the leads so that DNA will migrate from the wells toward the positive (red) electrode. **Remember: DNA runs toward the positive electrode — "run to red."**

13. Set the power supply to approximately 100 volts.

14. Run until the leading tracking dye has migrated about two-thirds to three-quarters of the way down the gel, typically 30 to 45 minutes.

15. Turn off and disconnect the power supply before opening the lid.

Part 4 — Visualize and document

16. Carefully transfer the gel to the transilluminator. Using the required eye protection and shield, photograph the band pattern.

17. Save and label the image with your sample identities and the date.

Expected Results

A successful gel shows:

- A **ladder lane** with a series of discrete, evenly spaced bands corresponding to the known fragment sizes printed on the ladder's reference chart.
- **Sample lanes** with one or more sharp, well-defined bands. To estimate the size of an unknown band, find the ladder bands immediately above and below it and interpolate.
- Bands that are crisp and roughly horizontal across the gel, not smeared or slanted.

To estimate a fragment's size, compare its migration distance to the ladder. A band that lines up between the 1,000 bp and 1,500 bp ladder bands is approximately 1,200 bp.

Troubleshooting

Observation	Likely cause	Corrective action
No bands visible at all	DNA degraded, too little DNA loaded, or stain omitted	Verify stain was added; load more DNA; check sample integrity
Smeared bands (no sharp lines)	DNA degraded, overloaded well, or run too hot/fast	Load less DNA; lower the voltage; use fresh sample
Bands run off the bottom of the gel	Gel run too long or voltage too high	Stop earlier; reduce voltage; watch the tracking dye

Wells torn or sample leaked out	Comb removed roughly, or loading dye omitted so sample floated away	Remove comb gently; always add loading dye before loading
Bands slanted or "smiling" across the gel	Uneven heat during the run, often from too high a voltage	Reduce voltage; ensure buffer fully and evenly covers the gel
Faint, fuzzy bands	Insufficient stain or weak transilluminator signal	Confirm stain concentration; check the imaging settings

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Companion Procedure: Setting Up a Restriction Digest

In many lab sessions, the DNA you run on the gel is the product of a **restriction digest** performed earlier in the period. Restriction enzymes (also called restriction endonucleases) cut double-stranded DNA at specific recognition sequences, typically four to eight base pairs long. Cutting a circular plasmid at a known number of sites produces a predictable set of fragment sizes, which the gel then reveals. This is why electrophoresis and restriction digestion are taught together: the digest creates the fragments, and the gel reads them out.

A typical 20 µL teaching-lab digest is assembled on ice in a microcentrifuge tube:

Component	Volume
Plasmid DNA (about 200 ng)	4 µL
10X restriction enzyme buffer	2 µL
Restriction enzyme	1 µL
Nuclease-free water	13 µL
Total	**20 µL**

Add the components in the order listed, keeping the enzyme on ice until the last moment and returning it to the freezer promptly — enzymes are heat-sensitive and lose activity at room temperature. Mix gently by flicking the tube, give it a brief spin in the microcentrifuge to collect the liquid, and incubate at the enzyme's optimal temperature (commonly 37 °C) for the time specified by the manufacturer, usually 1 hour. After incubation, add loading dye and the sample is ready for the gel.

Two predictable outcomes help students interpret their gels:

- An **uncut plasmid** control runs as supercoiled circular DNA, which migrates differently from linear fragments and often appears as one or two bands that do not match any linear-fragment size.
- A plasmid **cut once** linearizes and runs as a single band at the full plasmid size; a plasmid **cut twice** produces two bands whose sizes add up to the full plasmid length.

A Worked Interpretation Example

Suppose a 5,000 bp plasmid is digested with an enzyme expected to cut twice, producing 3,200 bp and 1,800 bp fragments. On the gel you would predict two bands: an upper band aligning near the 3,000 bp region of the ladder and a lower band near 1,800 bp. If instead you see a single band high on the gel, the enzyme likely cut only once (linearized) or failed to cut completely — a common result when the enzyme was left at room temperature too long or the incubation was too short. If you see a smear rather than discrete bands, the DNA may be degraded or the well overloaded. Reading the gel is therefore a test of both the technique and the underlying logic: the band pattern is a direct, physical readout of what the enzyme did to the DNA.

Tips for Clean Results

- Keep enzymes and DNA on ice; warmth is the enemy of both.
- Pipette accurately — at these small volumes, a 1 µL error is a 5 percent error in a 20 µL reaction.
- Load a ladder in the first lane every time; a gel without a size reference cannot be interpreted quantitatively.
- Do not overload wells. More DNA is not better; it smears the bands.
- Photograph promptly. Stained bands can diffuse and fade if the gel sits.

Buffer Chemistry: Why TAE Matters

The buffer is not an inert liquid; it carries the electric current and keeps the pH stable during the run. TAE (Tris-acetate-EDTA) maintains a slightly basic pH at which DNA's phosphate groups stay fully negatively charged, ensuring uniform migration toward the anode. The EDTA component chelates divalent metal ions such as magnesium, which would otherwise be available as cofactors for contaminating nucleases that degrade DNA. Running a gel without buffer, or with the wrong buffer, produces distorted or absent bands because the current has no consistent ionic path and the pH drifts.

Two practical rules follow from this chemistry. First, **the gel and the running buffer must be the same buffer at the same concentration.** A gel poured in 1X TAE but submerged in water, or in a different buffer, will run unevenly. Second, **old or repeatedly reused buffer loses its capacity.** After many runs the ions are depleted and the pH shifts, which shows up as slow migration, excessive heat, and fuzzy bands. When a gel mysteriously underperforms, fresh buffer is one of the first things to check.

Reading Gels Quantitatively

The relationship between fragment size and migration distance is approximately linear when migration distance is plotted against the logarithm of fragment size. In practice this means you can construct a rough standard curve from the ladder: measure how far each ladder band traveled, plot distance against the log of its known size, and read an unknown band's size off the line. For most teaching purposes, simple interpolation between the two flanking ladder bands is accurate enough, but understanding the log-linear relationship explains why the ladder bands bunch together at the top of the gel (large fragments, poorly resolved) and spread out toward the bottom (small fragments, well resolved). It also explains why choosing the right agarose percentage matters: a gel tuned to the wrong size range compresses the bands you care about into an unreadable cluster.

Glossary of Key Terms

- **Agarose:** A polysaccharide gel matrix that acts as a molecular sieve for DNA.
- **Anode / cathode:** The positive and negative electrodes; DNA migrates toward the anode (positive).
- **Base pair (bp):** The unit of DNA fragment length; one rung of the double helix.
- **DNA ladder:** A mixture of DNA fragments of known sizes used as a size reference.
- **Loading dye:** A dense, colored solution added to samples so they sink into wells and so progress can be tracked.
- **Restriction enzyme:** A protein that cuts DNA at a specific recognition sequence.
- **TAE buffer:** Tris-acetate-EDTA, the buffer that carries current and stabilizes pH.
- **Transilluminator:** A device that illuminates stained DNA for imaging, using UV or blue light.

Common Variations You May Encounter

Different courses and labs adapt this protocol in predictable ways, and recognizing the variation helps you adjust:

- **TBE buffer instead of TAE.** TBE (Tris-borate-EDTA) has higher buffering capacity and is preferred for long runs or small fragments, but it is less suitable if you intend to recover DNA from the gel afterward. The running principle is identical.
- **Higher or lower agarose percentage.** As the background table shows, a 2.0% gel is chosen when the fragments of interest are small (under a few hundred base pairs) and a 0.7% gel when they are large. The pouring and running steps are unchanged; only the resolution range shifts.
- **Pre-cast gels.** Many teaching labs now use commercially pre-cast gels with the stain already incorporated, which removes the molten-agarose and stain-handling steps but makes the buffer-matching and loading steps just as important.
- **Alternative stains.** Beyond SYBR Safe, some labs use other low-toxicity nucleic-acid stains. Always read the specific stain's safety data sheet; do not assume one stain's handling rules apply to another.

Whatever the variation, the core logic does not change: negatively charged DNA moves toward the positive electrode through a sieve, smaller fragments move faster, and a known ladder lets you turn migration distance into an estimated size.

Pre-Lab Questions

Answer these before coming to lab. You must be able to discuss them to be admitted to the bench.

1. In which direction does DNA migrate during electrophoresis, and why?
2. Why do smaller DNA fragments travel farther than larger ones in the gel?
3. Why must molten agarose be cooled to about 55 °C before the stain is added and the gel is poured?
4. State two specific safety hazards in this protocol and the precaution for each.
5. What is the purpose of the DNA ladder, and how would you use it to estimate the size of an unknown band?
6. What is the function of the loading dye, and what happens if you forget to add it?